

TINA REZVANIAN

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SUMMARY

Entrepreneurship & Innovation: Founder and CEO of Vinetka, leading vision, funding, and development of advanced AI systems. Created state-of-the-art applications in LLMs, VLMs, and multimodal learning, including Botlet (LLM-based smart shopping app) and DoD proposals on generative interoperability, ELINT, and video curation.

AI/ML Expertise: 7+ years building and deploying user-facing ML products. At Adobe, led development of generative models for Firefly and scalable experimentation pipelines. Skilled in generative modeling, RL, retrieval, and real-time personalization.

EDUCATION

Northeastern University Ph.D., Industrial Engineering and Operations Research Minor in Mathematics	Boston, MA 2013 - 2019
Northeastern Illinois University MBA, Master of Business Administration	Chicago, IL 2011 - 2013
University of Science and Technology of Mazandaran B.Sc., Industrial Systems Engineering	Mazandaran, Iran 2006 - 2010

SKILLS

AI/ML/DS: Vision and Language Architectures (Transformers, CLIP), Diffusion Models, Attention Mechanisms, VAEs, Temporal CNNs, Reinforcement Learning, Causal Models, Supervised Unsupervised Learning, Ensemble Methods, SVMs, Dimensionality Reduction, Clustering, Auto-Labeling, Graph Models, Community Detection, Normalization Regularization, Optimization, XGBoost, GBM.

Programming Python, PyTorch, scikit-learn, Spacy, SQL, Git, Hadoop, Spark, TensorFlow, Docker, Kubernetes, RunAI, CI/CD Pipelines, R, MATLAB, Flask, AWS, CPLEX, Gurobi, SQL, Hadoop.

Visualization Matplotlib, Plotly, Seaborn, ggplot2, Gephi, Tableau, Power BI.

WORK EXPERIENCE

Vinetka <i>Founder & CEO, Applied AI Research Lead</i>	Jun 2024 – Present <i>San Francisco, CA</i>
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- Founded and lead Vinetka, an applied AI company focused on the research, development, deployment, and commercialization of advanced machine learning systems and intelligent software platforms.
- Defined company vision and R&D strategy; led technical direction, proposal authorship, external engagement, and funding efforts for next-generation AI solutions powering complex technological systems and intelligent automation.
- Proposed and architected novel AI systems with applications in complex data environments, decision support, and information fusion:
 - . **BRIDGE:** A generative AI framework for harmonizing heterogeneous data sources and enabling model interoperability across systems.
 - . **TIDE:** A consensus-based interpretable learning method for detecting structured patterns across multiple input modalities.

- . **ANVIL**: An LLM-based platform for vulnerability identification and analysis in dynamic and unstructured data contexts.
- . **Video Curation**: A self-attention-driven model for real-time video summarization and compression based on semantic relevance.
- . **Signal Analysis**: Designed interpretable time-series models for robust detection and classification in noisy environments.
- Developed and launched **Botler**, an iOS app integrating multiple LLMs for deep search on natural language prompts and personalized in-app shopping recommendations.

Adobe

Sep 2019 - May 2024

Machine Learning Engineer, Data Scientist, Firefly

San Jose, CA

- Firefly: Developed and finetuned multiple Firefly's modules such as input/output data encoder, input dense vector retrieval, diffusion Unet, and in-painting inference
 - . Enriched data with generated captions, GenAI images, and multi-representational embeddings.
 - . Developed evaluation models to assess coherence and technical quality of images using human-scores
- AB Testing Firefly: Evaluated the impact of data enrichment strategies and architectural modifications to address model limitations and enhance performance.
 - . Developed a rapid experimentation framework automating training and evaluation; integrating RunAI, AWS, Artifactory; streamlining job configurations, checkpoints, Docker, GPU, and data transfer.
- Gap Analysis: Identified subjects, contexts, and actions where Gen AI performance was suboptimal.
 - . Resolved data issues in text presence, aesthetic scoring, face/human detection, and captioning.
 - . Extracted insights from unsupervised similarity, categorization, caption alignment, and segmentation.
 - . Optimized caption weights by NLP analysis, focusing on alignment, part of speech, and model biases.
- Multi-Channel Attention Attribution: Boosted paid user retention and trial conversion by recommending next-best action through estimating the credit share of each touchpoint and its time-of-use.
 - . Trained a scalable multi-head attention model with billions of parameters, visualized in a web app.
 - . Identified critical workflows for engagement during the user journey
- Super Users: Segmented and characterized users to recommend workflow sequences, optimizing retention and personalizing onboarding, in-app, and email communications.
 - . Trained a β -variational auto-encoder to automatically extract meaningful attributes
 - . Generated similar sequences to that of retained users using the decoder within attribute-based clusters
- FontMART: Ranked fonts to maximize reading speed using LambdaMART, directly optimizing NDCG.
 - . Gathered and evaluated data from MTurk, web apps, and Hadoop to ensure quality and readiness.
 - . Identified optimal fonts, fastest speeds, and key font traits by age group.
- RealTime Recommender: Developed an on-device real-time recommender using reinforcement learning, trained via interactions with a generative AI user agent.
 - . Trained a generative adversarial network to model generalized user behavior from sequential data.
 - . Built a general deep Q-learning recommender and personalized it through user-specific fine-tuning.
- CausalTIM: Performed causal analysis to identify features' impact on conversion using observational data.
 - . Developed a parallel workflow for propensity score matching across features, handling data preparation, instance matching, and treatment effect estimation.

Harvard Extension School

Nov 2019 - May 2020

- Face Recognition: Developed face recognition models with CNNs to match extracted faces to identities in the CASIA-WebFace dataset.
 - . Adapted FaceNet, DeepFace, VGGFace, and ResNet models using feature extraction, transfer learning, fine-tuning, data augmentation, and regularization

Insight Data Science

Jan 2019 - Feb 2019

- PickaBasket: Developed a web application with Python, Flask, and AWS to offer bestseller bundles for online grocery stores using graph algorithms.
 - . Identified patterns in customers' purchase behavior as probabilistic rules on bestseller combinations.

Wheels UP Private Jet Company

May 2018 - Aug 2018

Data Science Intern

New York City, NY

- Forecasted daily flight demand and allocated aircraft to regions to minimize empty flights, clustering movements with regional metrics and community detection methods.

Northeastern University

Dec 2014 - Sep 2019

Research Assistant

Boston, MA

- Forecasted emergency response demand for refugees at UNHCR with a modified regression gradient boosting Tree (GBM), outperforming Tweedie and Poisson models for zero-inflated skewed variables.
- Developed a matching algorithm with negotiated Top Trading Cycle for multi-period markets on shortlists at the World Food Programme, delivering a Python-based decision support web app.
- Integrated emergency and ongoing supply chains in a multi-objective framework, reducing cost by 31% and response time by 18% with a two-stage stochastic model, delivering a Python-based decision support.

Healthcare Systems Engineering Institute (HSYE)

Sep 2013 - Dec 2014

Northeastern University, Research Assistant

Boston, MA

- Scheduled Treatments and Nurses using Column Generation Algorithm at Moffit Cancer Center.
- Optimized Curbside Threshold at Massachusetts General Hospital (MGH) Neurology Department.

PUBLICATIONS

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- . Beier, Sofie, et al. "Readability research: An interdisciplinary approach." *Foundations and Trends® in Human-Computer Interaction* 16.4 (2022): 214-324.
 - . Cai, T., Wallace, S., **Rezvanian, T.** (2022, June). Personalized Font Recommendations: Combining ML and Typographic Guidelines to Optimize Readability. In *Designing Interactive Systems Conference* (pp. 1-25)
 - . Jahre, M., Kembro, J., **Rezvanian, T.**, Ergun, O., Haapnes (2016) *Integrating supply chains for emergencies and ongoing operations in UNHCR*, *Journal of Operations Management*, 45, 57-72.
 - . Jahre, M., Kembro, J., **Rezvanian, T.**, Ergun, O., Haapnes, S., & Hultberg, M. *Scenario building in UNHCR-Predicting future demand in refugee crises*. In *Proceedings of the 5th World Production and Operations Management Conference, P&OM Havana 2016*
 - . **Rezvanian, T.**, Ergun, O., Jahre, M., Kembro, J., (2020) *Forecasting Emergency Response Demand for Refugees as a Zero-Inflated Skewed variables: Evidence from UNHCR*
 - . **Rezvanian, T.**, Ergun, O., *Two-sided Stable Matching with Negotiations*, *Integrating Data-Driven Forecasting and Large-Scale Optimization to Improve Humanitarian Response Planning and Preparedness*. Chapter 1, Diss. Northeastern University,
 - . **Rezvanian, T.**, Ergun, O., *Deferred Acceptance Algorithms and Aftermarkets for Matching with Short Preference Lists at World Food Programme*, *Integrating Data-Driven Forecasting and Large-Scale Optimization to Improve Humanitarian Response Planning and Preparedness*. Chapter 1, Diss. Northeastern University, 2019.